

ICT 3101 — Operating System

Chapter 12: Mass-Storage Systems

Exam Revision Note

1. Overview of Storage

1.1 Primary Memory

- Primary memory is used directly by the processor while the computer is running.
- It holds programs and data that are being used right now.
- It is volatile. This means data is lost when power is off.
- RAM is the most common example.
- When a computer starts, the OS, user interface, and running programs are loaded into main memory.

1.2 Secondary Memory

- Secondary memory is non-volatile. It keeps data even when the computer is off.
- The processor cannot use it directly. Data first goes to main memory, then to the processor.
- Examples: ROM, flash drives, hard disk drives (HDD), magnetic tapes.
- It is slower than primary memory but has much bigger storage size (MB to TB).
- All mass storage devices belong to secondary memory.

1.3 Primary Memory vs Secondary Memory

Point	Primary Memory	Secondary Memory
Access	Used directly by processor	Not accessed directly by processor
Volatility	Volatile (data lost on power off)	Non-volatile (data stays after power off)
Speed	Fast	Slower
Size	Small size	Large size (MB to TB)
Example	RAM	HDD, ROM, flash drive, magnetic tape

2. Disk Scheduling

Disk scheduling (also called I/O scheduling) is done by the OS to decide the order of I/O requests sent to the disk.

Why disk scheduling is needed:

- Many I/O requests come from different processes, but the disk can serve only one at a time. Others must wait in a queue.
- Requests may be far apart on the disk. This causes large arm movement if not scheduled well.
- Hard drives are slow parts of the computer, so they must be used efficiently.

2.1 Purpose and Goals

The disk scheduling algorithm picks a request from the queue and decides when it will be processed.

Goals:

- Fairness
- High throughput
- Minimal head travel time

2.2 Disk Access Time (Formula)

Disk Access Time = Seek Time + Rotational Latency + Transfer Time

- **Seek Time** → Time needed to move the disk head to the correct track.
- **Rotational Latency** → Time needed for the required sector to rotate under the head.
- **Transfer Time** → Time needed to actually read or write the data.

Disk Response Time = Average time a request waits for its I/O operation to finish.

Full path from queue to finish: Queuing Delay → Seek Time → Rotational Latency → Transfer Time (together = Disk Access Time).

3. Disk Scheduling Algorithms

Each request list example below uses: Requests = 82, 170, 43, 140, 24, 16, 190. Current head position = 50.

3.1 FCFS (First Come First Served)

Meaning: Requests are served in the exact order they arrive in the queue.

How it works:

- The disk head moves to each request one by one, in arrival order.

Advantages:

- Every request gets a fair chance.
- No indefinite postponement (no starvation).

Disadvantages:

- Does not try to reduce seek time.
- May not give the best performance.

Example: Order served: 82, 170, 43, 140, 24, 16, 190. Total seek time = 642.

3.2 SSTF (Shortest Seek Time First)

Meaning: The request nearest to the current head position is served first.

How it works:

- Seek time of every waiting request is calculated.
- The request with the smallest seek time is served next.

Advantages:

- Average response time decreases.
- Throughput increases.

Disadvantages:

- Extra work needed to calculate seek time in advance.
- Can cause starvation for far away requests.
- High variance in response time because it favors nearby requests.

Example: Order served: 43, 24, 16, 82, 140, 170, 190. Total seek time = 208.

3.3 SCAN

Meaning: The disk head moves in one direction, serving requests on the way. At the end of the disk, it reverses direction and serves requests on the way back. It works like an elevator, so it is also called the elevator algorithm.

How it works:

- Head moves towards one end (say, larger track numbers), serving requests in its path.
- On reaching the last track, it turns around.
- It then serves remaining requests while moving back.

Advantages:

- High throughput.
- Low variance of response time.
- Good average response time.

Disadvantages:

- Requests just visited by the head wait a long time before the head returns.

Example: Head moves 50 → 199, then 199 → 16. Total seek time = 332.

3.4 C-SCAN (Circular SCAN)

Meaning: Like SCAN, but instead of reversing direction at the end, the head jumps straight to the other end of the disk and starts serving requests again from there. This makes the arm movement circular.

How it works:

- Head moves in one direction, serving requests.
- At the end, it jumps to the start of the disk (no serving during the jump).
- It then serves requests moving in the same direction as before.

Advantages:

- Gives more uniform (even) waiting time compared to SCAN.

Disadvantages:

- Extra time used to travel back to the start without serving any request.

Example: Head moves 50 → 199, jumps to 0, then 0 → 43. Total seek time = 391.

3.5 LOOK

Meaning: Similar to SCAN, but the head does not go all the way to the end of the disk. It only goes as far as the last request in that direction, then reverses. This avoids wasted movement.

How it works:

- Head moves towards pending requests only, not to the physical end of the disk.
- When no more requests are ahead, it reverses direction.
- It then serves requests on the way back.

Advantages:

- Gives more uniform waiting time compared to SCAN.
- Avoids unnecessary travel to the disk end.

Disadvantages:

- Still less fair to newly arriving requests behind the head, similar to SCAN's general pattern.

Example: Head moves 50 → 190 (last request), then 190 → 16. Total seek time = 314.

3.6 C-LOOK

Meaning: Similar to C-SCAN, but the head only goes as far as the last request in each direction, not to the physical end of the disk, before jumping to the other side.

How it works:

- Head moves towards the last pending request in one direction.
- It then jumps directly to the last pending request on the other side.
- It continues serving requests from there in the same direction as before.

Advantages:

- Avoids unnecessary travel to the disk end, like LOOK.
- Gives uniform waiting time, like C-SCAN.

Disadvantages:

- Extra jump time to the other side, though shorter than C-SCAN's jump.

Example: Head moves 50 → 190, jumps to 16, then 16 → 43. Total seek time = 341.

3.7 Comparison of Disk Scheduling Algorithms

Algorithm	Main Idea	Advantage	Disadvantage
FCFS	Serves requests in arrival order	Simple, fair	High seek time, not optimized
SSTF	Serves nearest request first	Less average seek time	May cause starvation
SCAN	Moves one direction, then reverses (elevator)	High throughput, low variance	Recently visited areas wait long
C-SCAN	Moves one direction, jumps back to start	Uniform wait time	Wasted return jump time
LOOK	Like SCAN but stops at last request	No wasted travel to disk end	Similar fairness issue as SCAN
C-LOOK	Like C-SCAN but stops at last request	No wasted travel, uniform wait	Still has a jump between ends

3.8 SCAN vs C-SCAN

Point	SCAN	C-SCAN
Direction change	Reverses direction at the end	Jumps to the other end, same direction
Servicing while returning	Serves requests while going back	Does not serve while jumping back
Wait time	Less uniform	More uniform

3.9 LOOK vs C-LOOK

Point	LOOK	C-LOOK
Based on	SCAN (goes only to last request)	C-SCAN (goes only to last request)
Direction change	Reverses at last request	Jumps to last request on other end
Travel to disk end	Avoided	Avoided

3.10 Selecting a Disk Scheduling Algorithm

- SSTF is common and has a natural appeal.
- SCAN and C-SCAN work better for systems with heavy disk load, because they cause less starvation.
- Performance depends on the number and type of requests.
- The scheduling algorithm should be a separate OS module, so it can be replaced easily.
- SSTF or LOOK is a reasonable default choice.

4. Overview of Mass Storage Structure

- Mass storage devices store very large amounts of data.
- The main purpose of mass storage is data backup and data recovery.
- Types of mass storage devices: Magnetic Disks, Solid State Disks, Magnetic Tapes.

4.1 Magnetic Disks

- A magnetic disk stores data using magnetization. Data is written, rewritten, and read this way.
- It has tracks, sectors, and spots for storing data.
- IBM made the first magnetic hard drive in 1956. It could store only 5 MB.
- Common types: Floppy Disks, Hard Disks, Zip Disks.

Structure and Working

- A mechanical arm moves across a spinning platter (the disk surface). This is called a comb.

- Platters are round and coated with magnetic material. Hard disk platters are made of stiff metal.
- A read-write head floats just above the platter on a thin layer of air.
- Small current magnetizes tiny spots on the disk to write data, and reads the magnetized spots to read data.
- Data is stored in tracks (circles) and sectors (parts of a track). Each sector usually stores 512 bytes.
- A cylinder is a group of tracks that are at the same distance from the edge, across all platters.
- If the head touches the disk by accident, it is called a head crash. This can damage the disk.
- Heads are 'parked' (moved off the disk) when the computer shuts down, to avoid damage.

Interfaces

Disks connect to computers through an I/O Bus. Common interface types: EIDE, ATA, SATA, USB, Fiber Channel (FC), SCSI.

The disk controller (inside the disk) moves data between the magnetic surface and its cache. The host controller (on the computer side) talks to the CPU.

4.2 Disk Structure

- A disk is treated as a large 1-dimensional array of logical blocks. The logical block is the smallest unit that can be transferred.
- Low-level formatting creates these logical blocks on the physical disk.
- Sector 0 is the first sector of the first track on the outermost cylinder.
- Mapping goes track by track, then cylinder by cylinder, from outermost to innermost.
- Some sectors may be bad, which can break the simple logical-to-physical mapping.

4.3 Magnetic Disk Performance Terms and Formula

- **Seek Time** → Time to move the disk arm to the required track.
- **Rotational Latency** → Time for the required sector to rotate to the read-write head.
- **Transfer Time** → Time to actually transfer the data.

Disk Access Time = Seek Time + Rotational Latency + Transfer Time

- **Disk Response Time** → Average time a request waits for its I/O to complete.

Extra performance facts (from the chapter):

- Platters range from 0.85" to 14" historically; common sizes are 3.5", 2.5", 1.8".
- Disk size ranges from 30 GB to 3 TB per drive.
- Theoretical transfer rate: 6 Gb/sec. Real (effective) transfer rate: about 1 Gb/sec.
- Seek time ranges from 3 ms to 12 ms; 9 ms is common for desktop drives.
- Average seek time is often calculated using 1/3 of the tracks.
- Latency formula: $1 / (\text{RPM} / 60) = 60 / \text{RPM}$.
- Average latency = half of the full rotation latency.

4.4 Magnetic Tapes

- Used before hard disks became common; now mostly used for backups.
- Finding a specific spot on tape can take time, but once started, reading/writing speed is close to disk drives.
- Tape capacity ranges from 20 GB to 200 GB. Compression can double this capacity.

4.5 Solid State Disks (SSD)

- SSDs act like a small, fast hard disk, but use memory chips (flash memory or battery-backed DRAM) instead of spinning platters.
- SSDs have no moving parts, so they are much faster than normal hard disks.
- Because there are no moving parts, disk-scheduling problems (like seek time) do not apply to SSDs.

- SSDs are used as a high-speed cache for data that must be fetched quickly, such as file system metadata (directory and inode information).
- SSDs are also used as boot drives (holding OS and some programs) and in laptops to make them thinner and faster.
- Since SSDs are very fast, they are sometimes connected directly to the PCI bus to avoid bus speed limits.

4.6 HDD vs SSD

Point	HDD (Magnetic Disk)	SSD (Solid State Disk)
Moving parts	Has moving parts (arm, spinning platter)	No moving parts
Speed	Slower	Much faster
Disk scheduling needed?	Yes	Not needed (no seek time issue)
Technology	Magnetization on platters	Flash memory or battery-backed DRAM
Common use	General storage, backup	Cache, boot drive, laptops